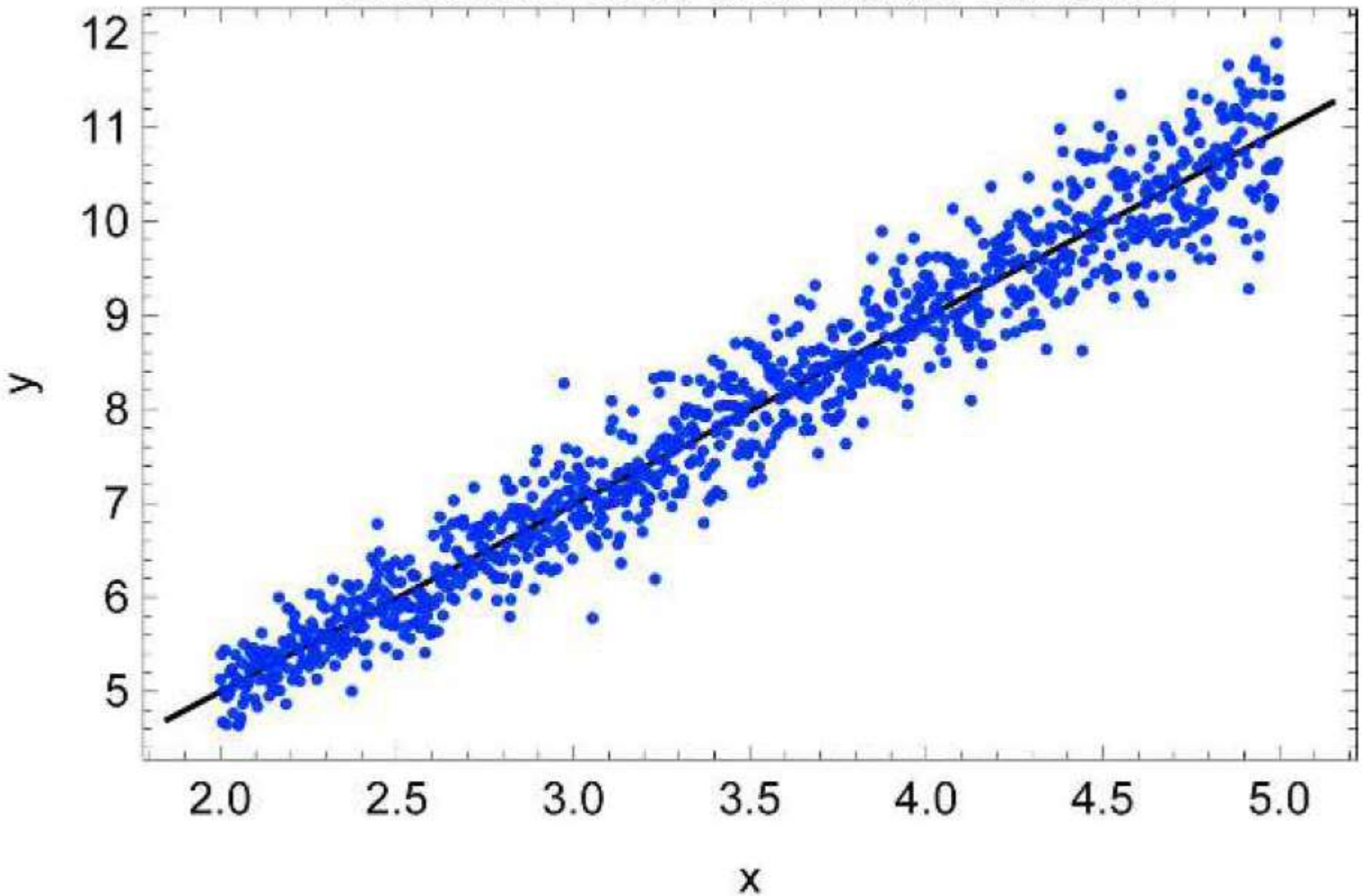
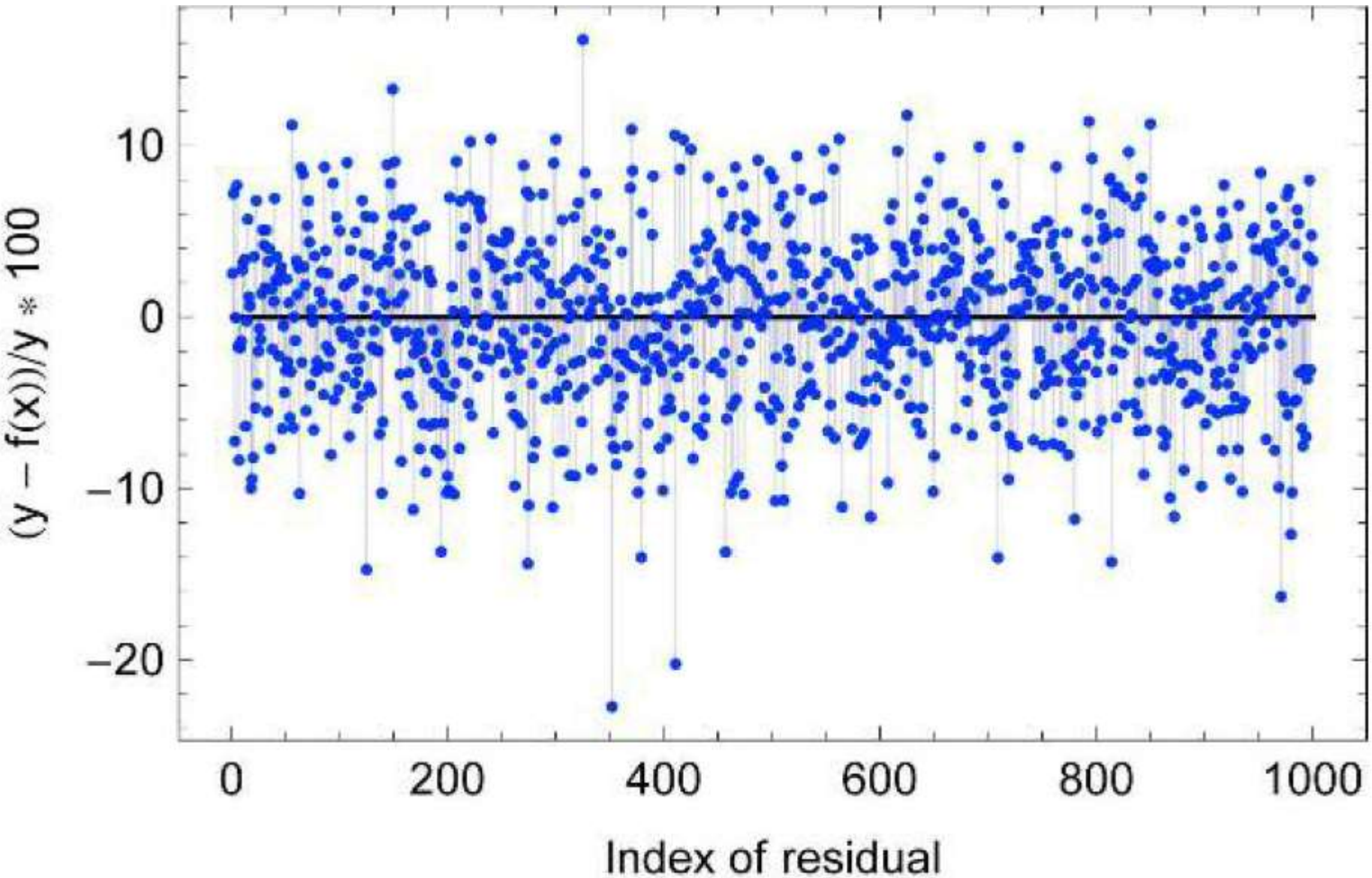
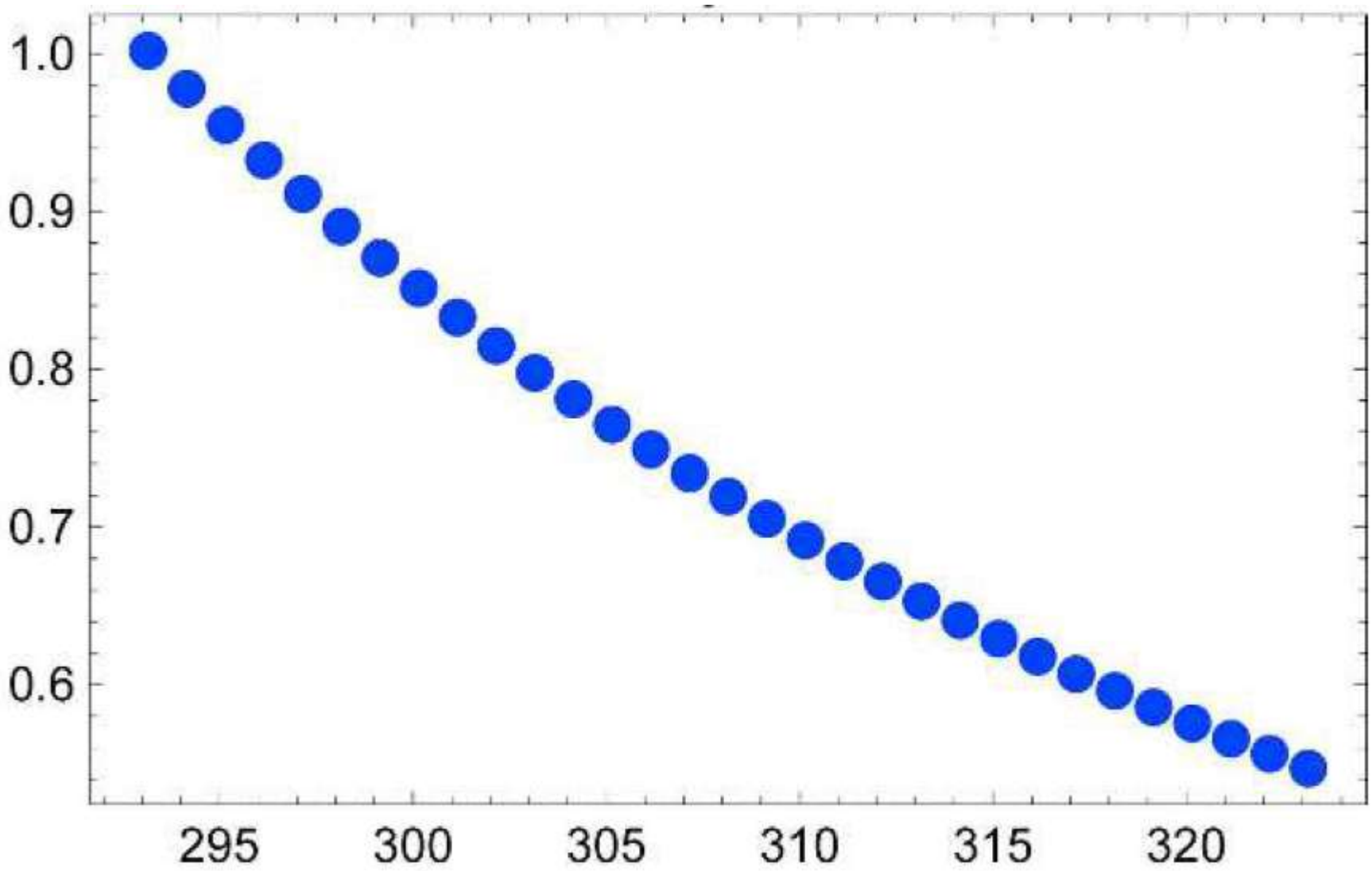
# Least Mean Square Error

R Srinivasan
Faculty
Department of Information Technology
SSN College of engineering
OMR, Kalavakkam-603 110

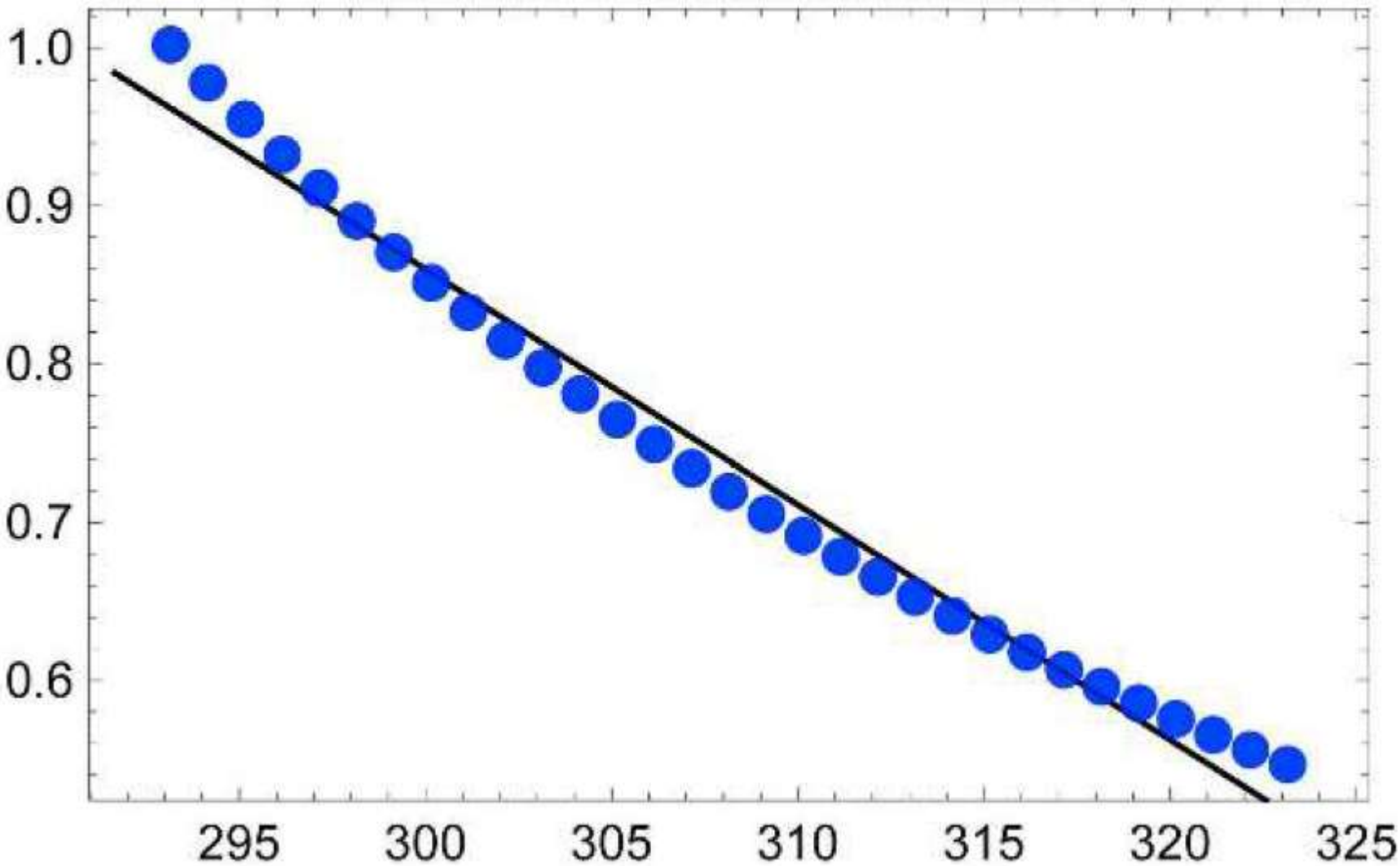

12
11
10
9
8
7
6
5
y
2.0
2.5
3.0
3.5
4.0
4.5
5.0
x


10
0
−10
−20
(y − f(x))/y * 100
0
200
400
600
800
1000
Index of residual

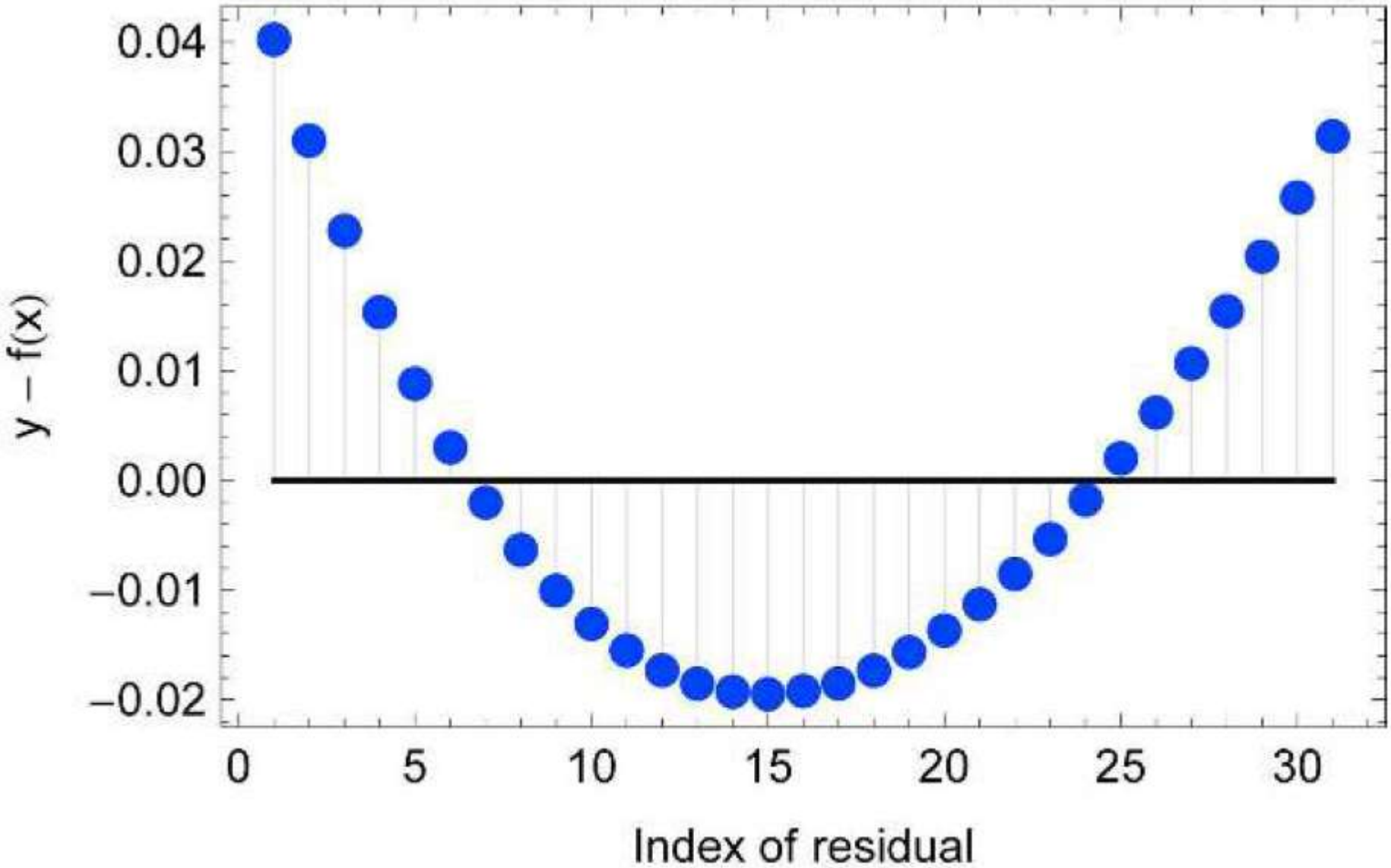
# Given data



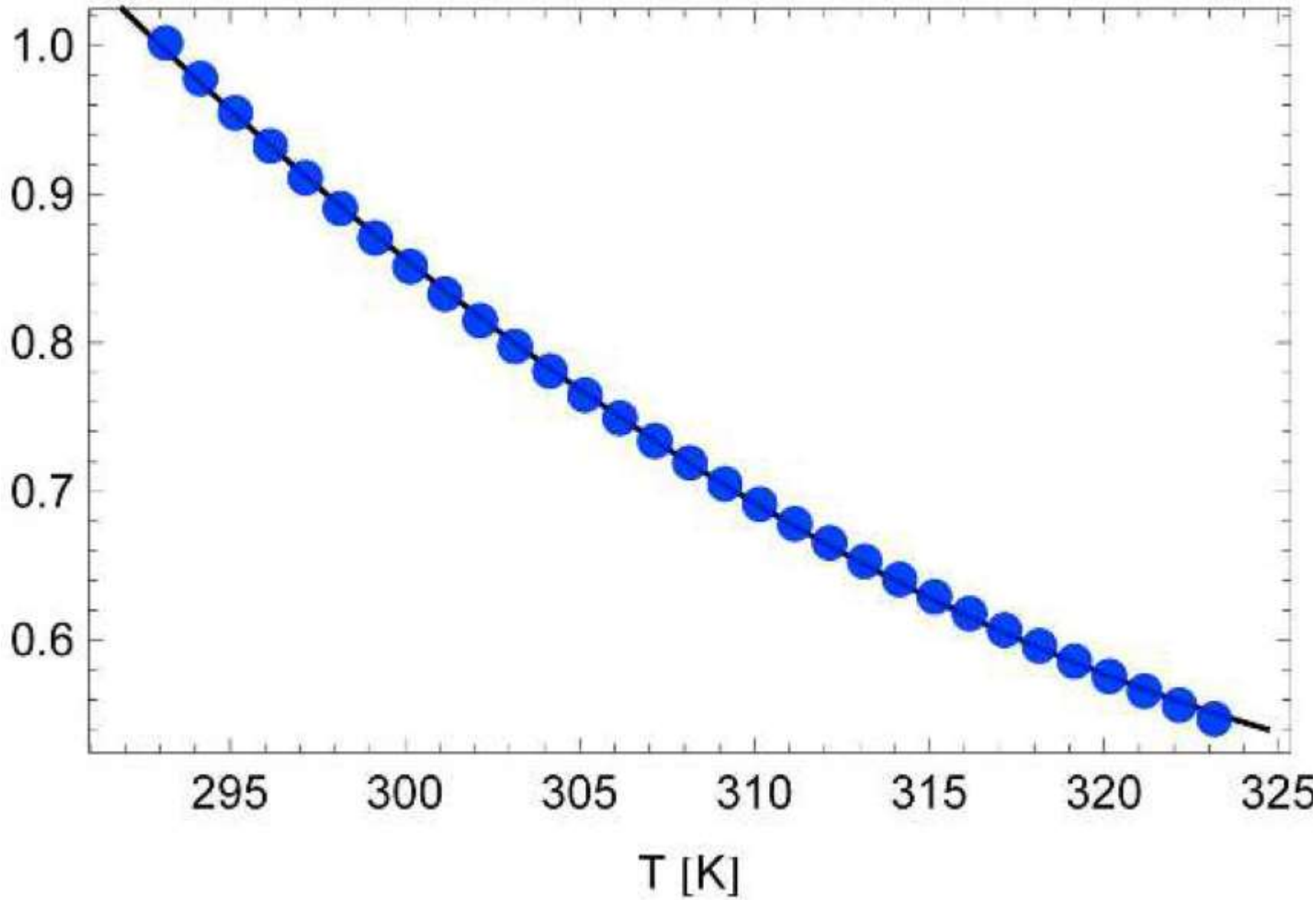
# Assume a function to fit $y = a_1 + a_2x$



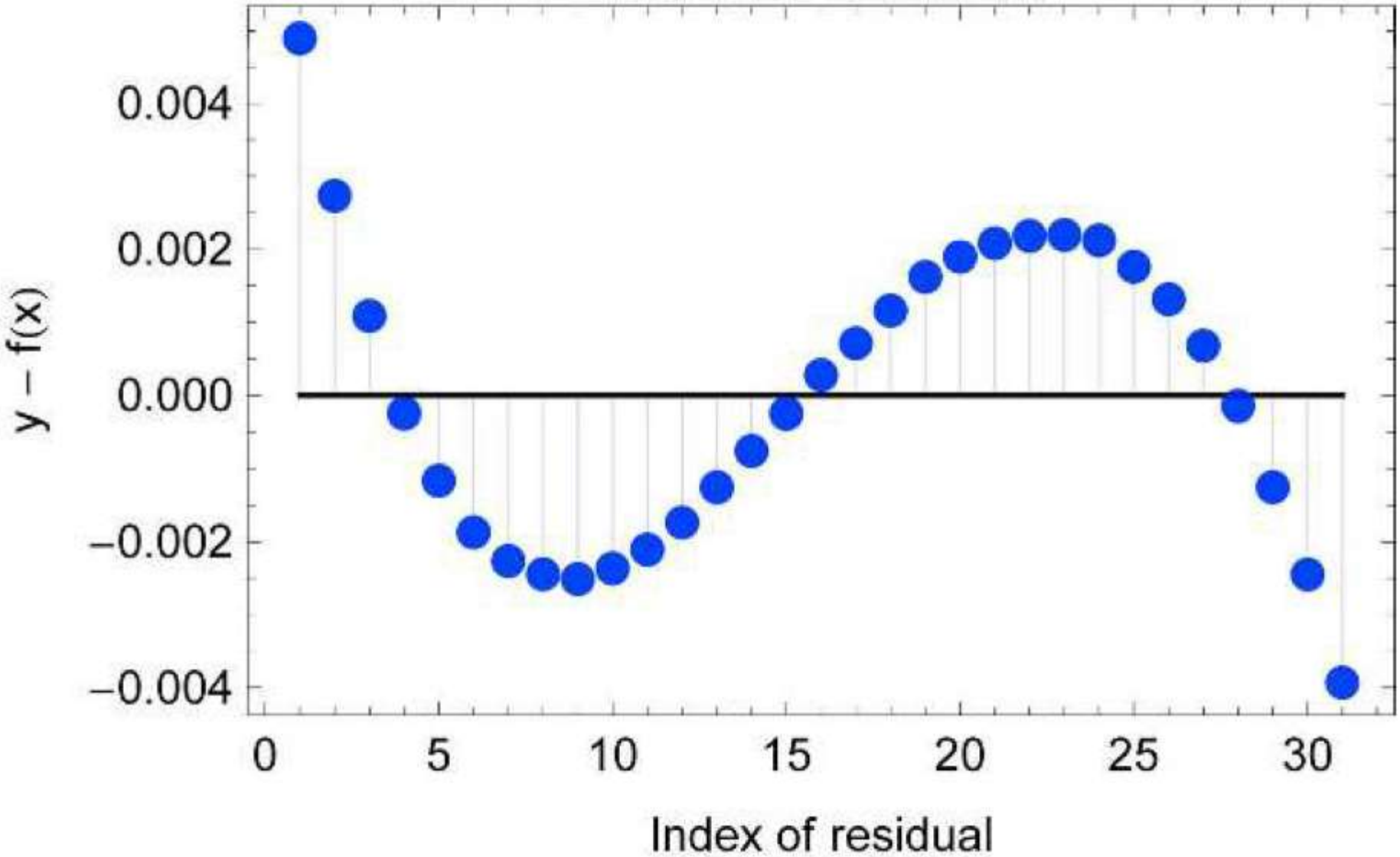
# Residual or errors @ various points



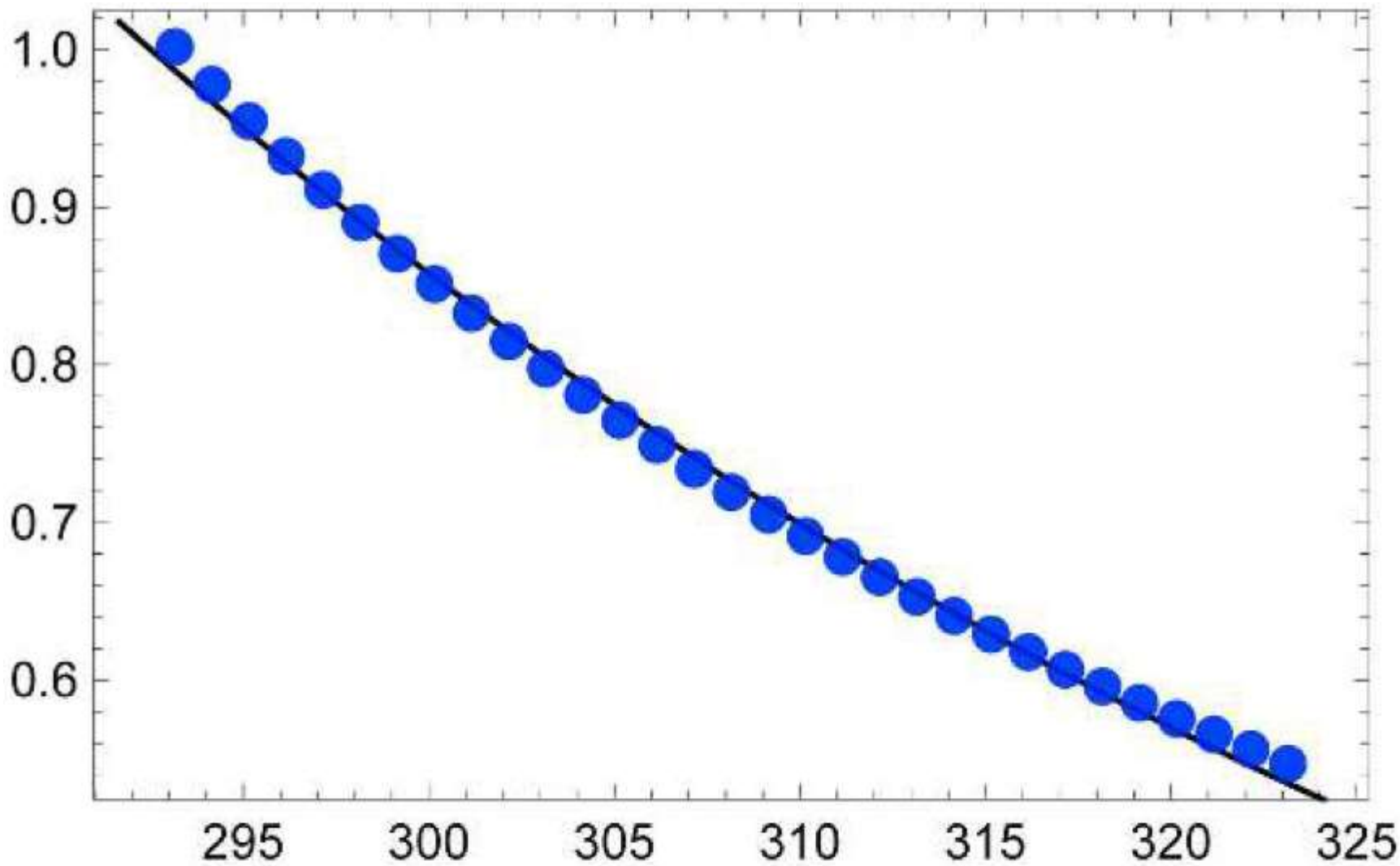
# Assume function to fit $y = a_1 + a_2x + a_3x^2$



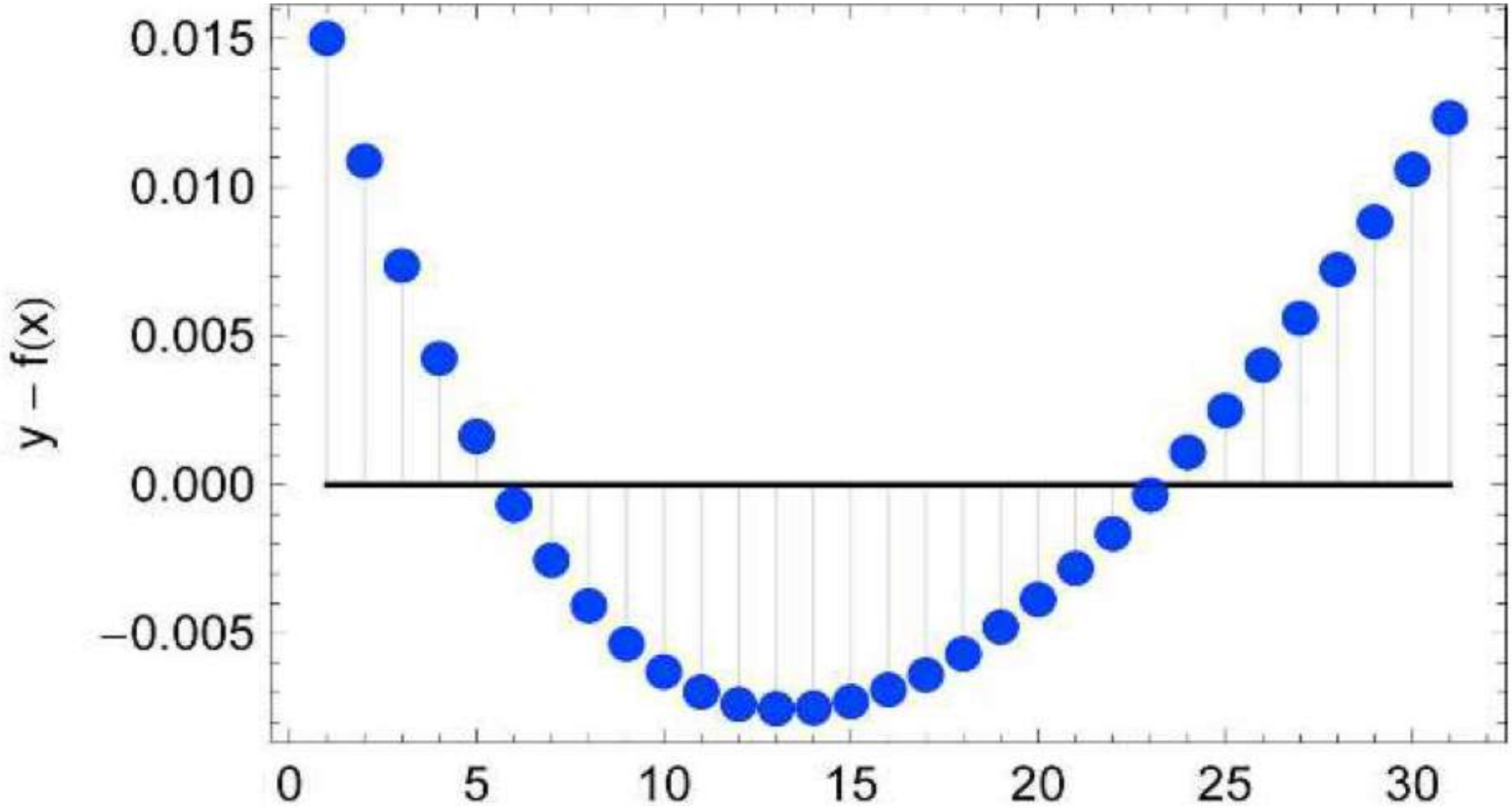
# Residual or errors @ various points



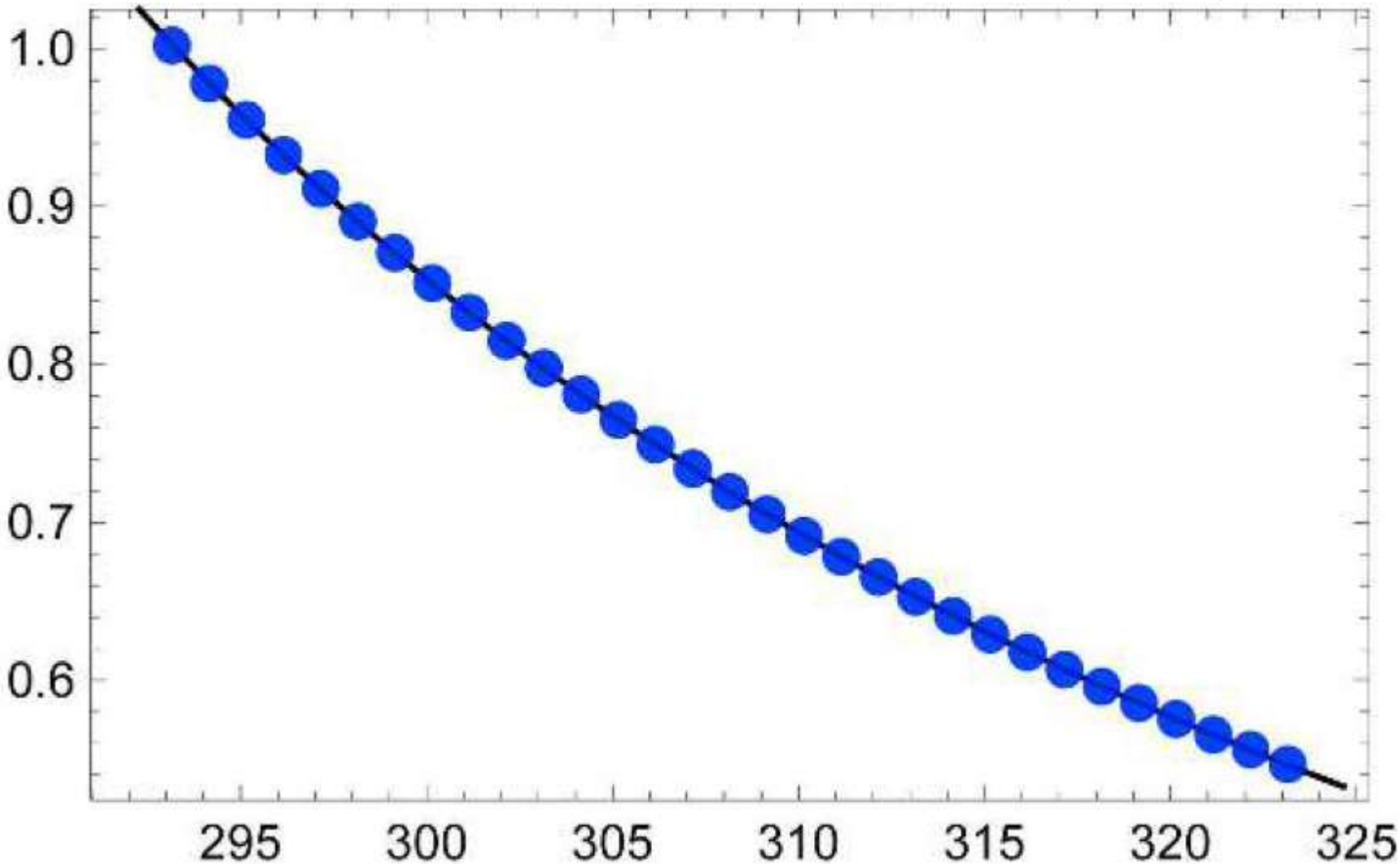
Assume function, $y = a_1 . e^{a_2 x}$



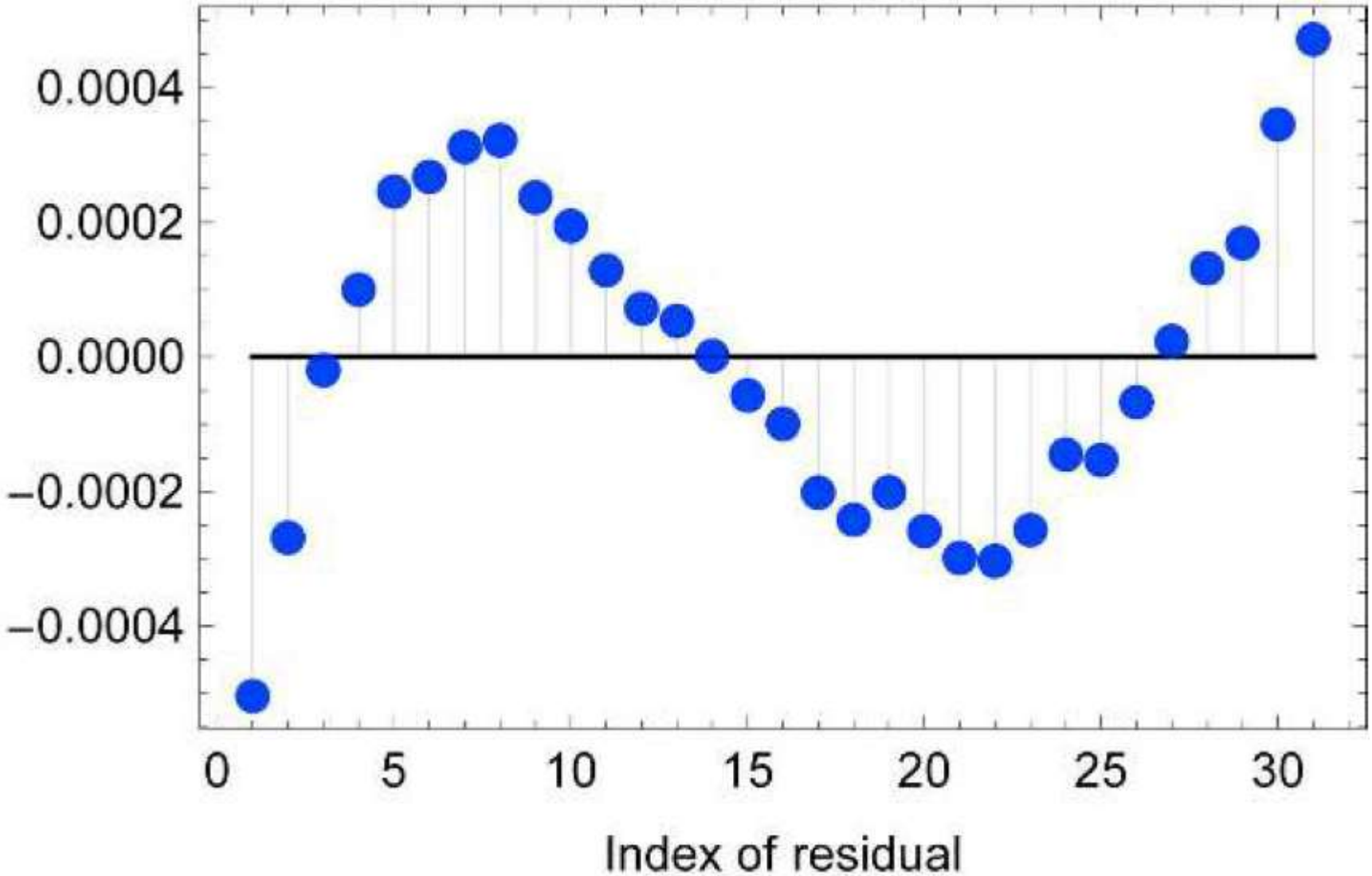
# Residual or errors @ various points



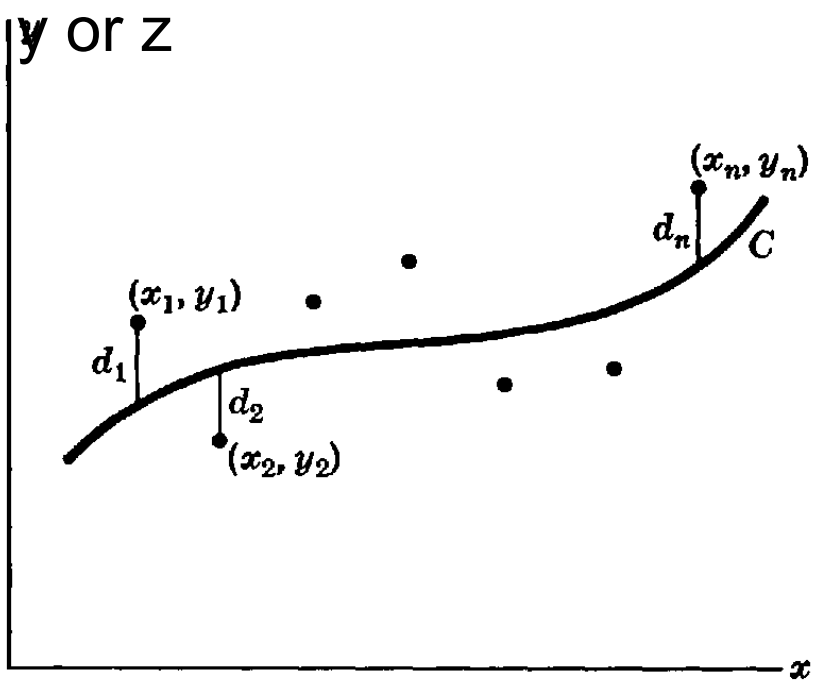
Assume function to fit , $y = a_1 + \frac{a_2}{(a_3 - x)}$



# Residual or errors @ various points



# Deviation - error




- Given, $x_i$ and $y_i$
- K number of data points
- Error = $y_i - f(x_i)$
- How many errors?
- K errors

Sum of Squared errors

$$d_1^2 + d_2^2 + \cdots + d_k^2 = \text{a minimum}$$

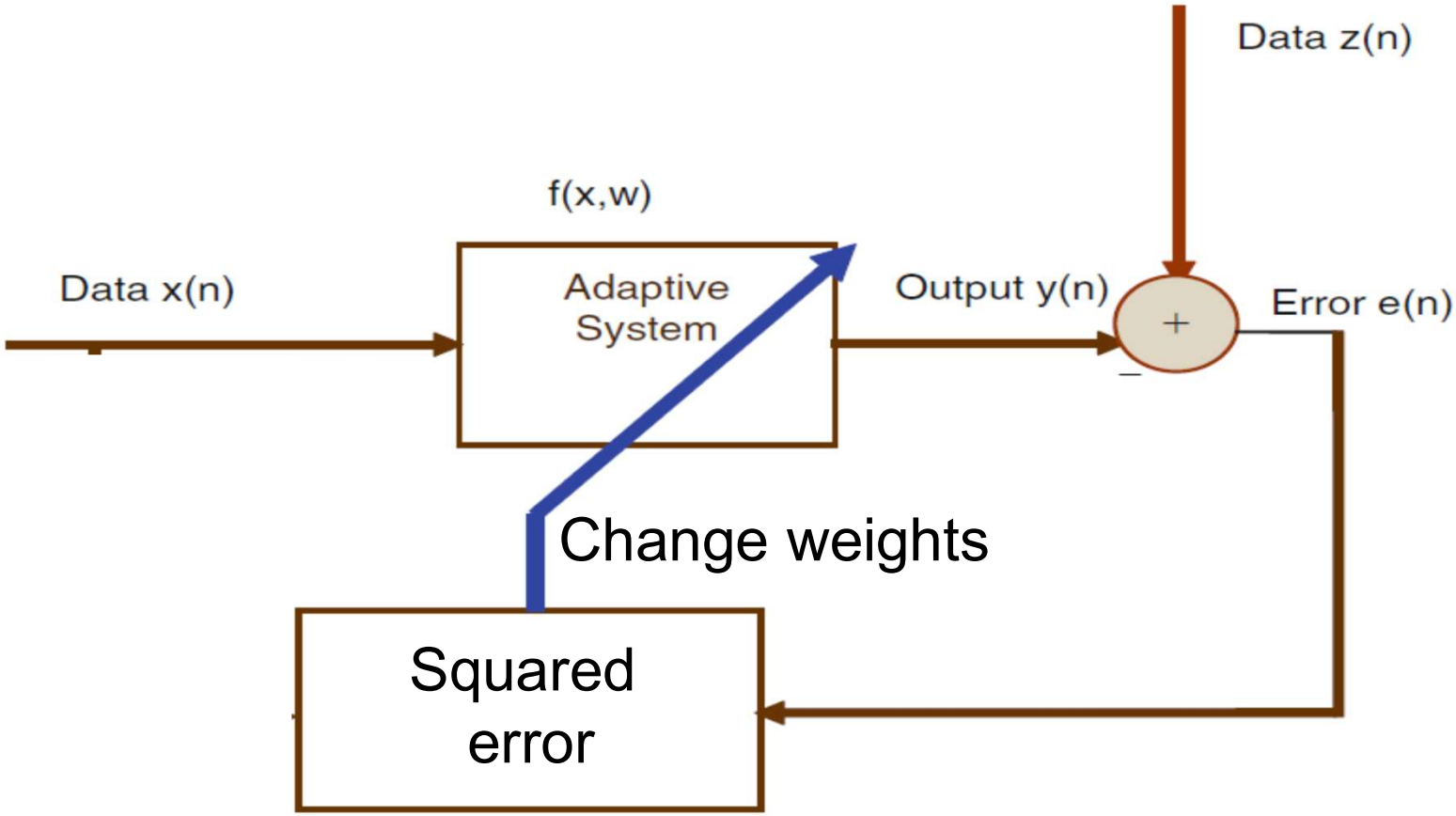
$$\sum_{i=1}^{K} (y_i - f(x_i))^2 \longrightarrow \text{minimize}$$

# Root mean square error

$$\mathrm{RMSE} = \sqrt{\frac{1}{K}\sum_{i=1}^{K}\left(y_i - f\left(x_i\right)\right)^2}$$

# Least MSE - Schematic

Squared error = $[(z(n) - f(\mathbf{w}, x(n)))^2]$

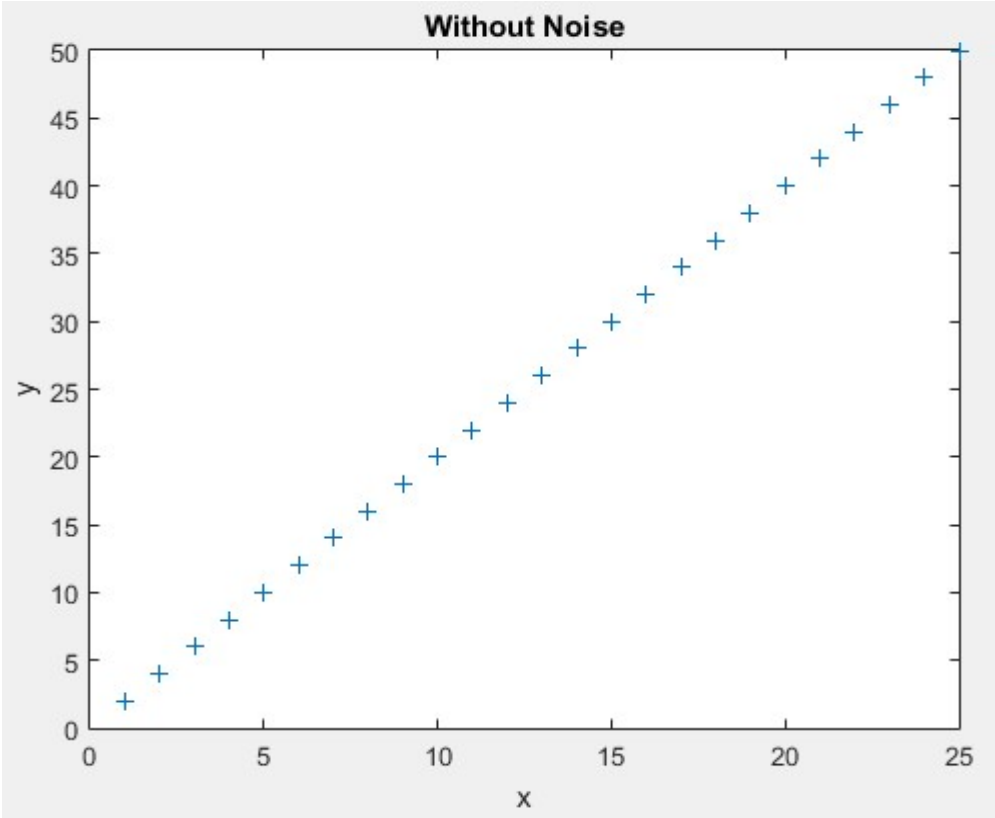


# Data points generated: t=2x

| X | y |
|---|---|
| 1 | 2 |
| 2 | 4 |
| 3 | 6 |
| 4 | 8 |
| 5 | 10 |
| 6 | 12 |
| 7 | 14 |
| 8 | 16 |
| 9 | 18 |
| 10 | 20 |
| 11 | 22 |
| 12 | 24 |
| 13 | 26 |
| 14 | 28 |

| x | y |
|---|---|
| 15 | 30 |
| 16 | 32 |
| 17 | 34 |
| 18 | 36 |
| 19 | 38 |
| 20 | 40 |
| 21 | 42 |
| 22 | 44 |
| 23 | 46 |
| 24 | 48 |
| 25 | 50 |

# t=2x points plotted

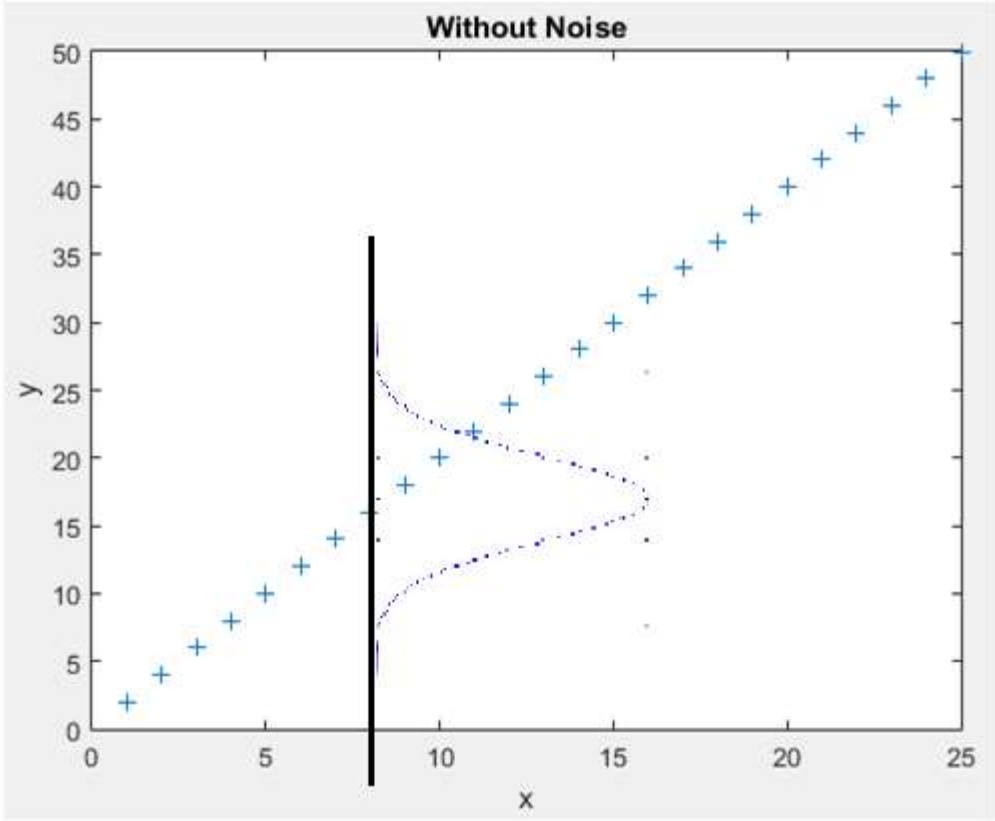



### 🧠 Conceptual summary

| Concept | Mathematical meaning | Role |
|---|---|---|
| **Error** | $e_i = y_i - f(x_i)$ | deviation between predicted and true values |
| **SSE / MSE** | $\sum e_i^2 / K$ | quantitative measure of fit quality |
| **Least-Squares principle** | choose parameters minimizing MSE | foundation of LMS algorithm |
| **Cost function J(w)** | $E[(z - w^T x)^2]$ | function of weights, minimized in learning |
| **Adaptive system** | updates w to minimize J over time | forms basis of adaptive filters |
| **Orthogonality condition** | $E[xe] = 0$ at optimum | implies minimum MSE reached |

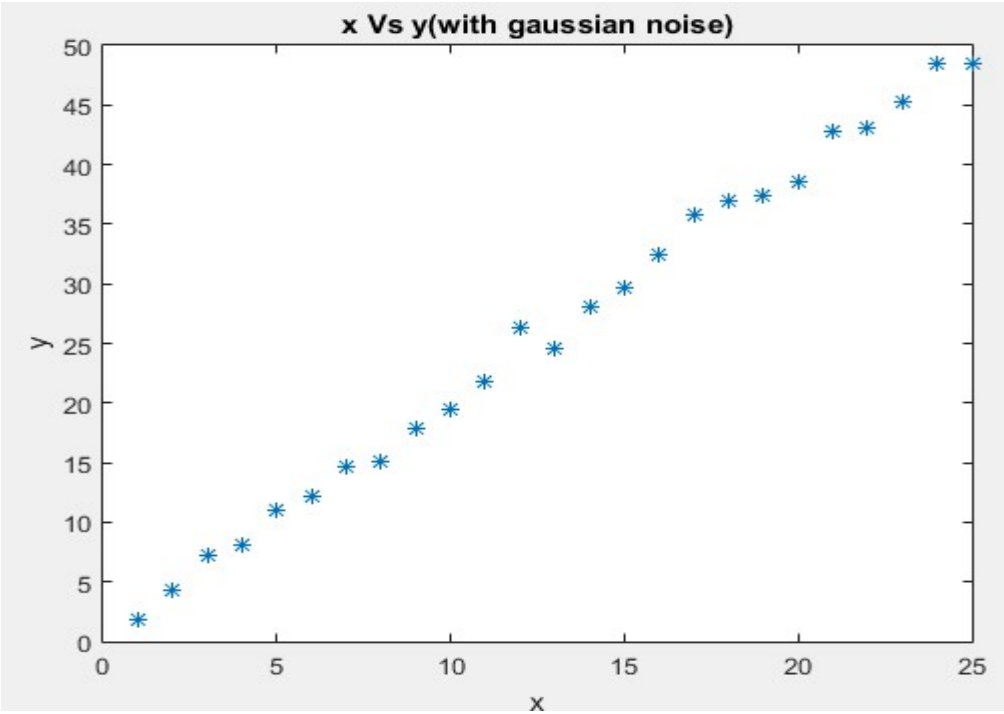
### 🏁 Overall takeaway

1. **Least Mean Square (LMS)** is the practical, adaptive implementation of the **Least Squares** principle.
2. It minimizes the **mean square of the error signal** → best average fit to noisy data.
3. In information-theoretic terms, it reduces uncertainty (entropy of error) and aligns model output with the information in the data.
4. This idea underlies adaptive filters, prediction, and many learning algorithms.


Without Noise
50
45
40
35
30
25
20
15
10
5
0
y
0
5
10
15
20
25
x

# t=2x + Gaussian noise ($\sigma$=2)

All the points are disturbed



# Algorithm

- We have decided to fit using y=m.x
- x=[---, ---, ---, ---, …., ---]
- t=[---, ---, ---, ---, …., ---]
- Choose m

***Predict y=[---, ---, ---, ---, …., ---]***

**<u>Find out error</u>**

***e=[---, ---, ---, ---, …., ---]***

**<u>Generate squared error by squaring the elements of e</u>**

***se=[---, ---, ---, ---, …., ---]***

**<u>Compute the mean of se (MSE)</u>**

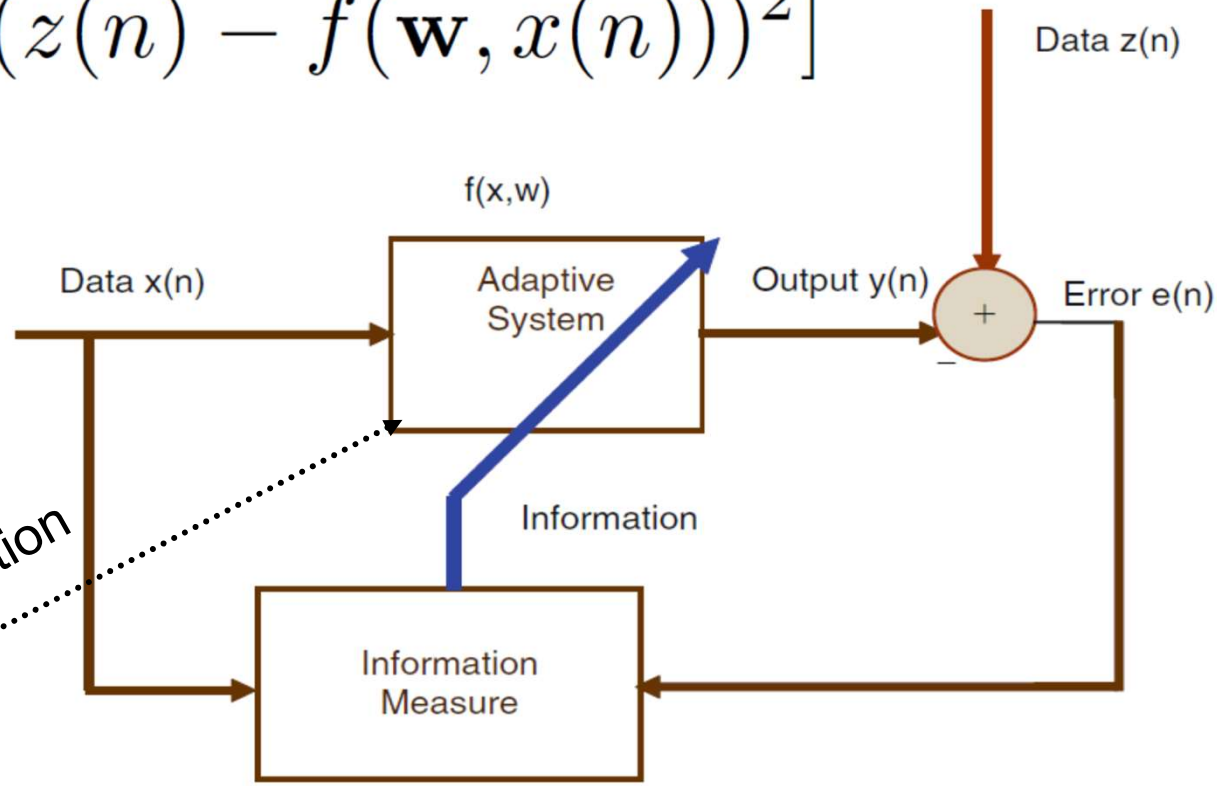
***MSE = $\sum$ se***

- Change m; repeat 1 to 6
- Plot MSE versus m; choose the m corresponding to minimum MSE

# MSE for a linear system

$$J_w(e(n)) = E[(z(n) - f(\mathbf{w}, x(n)))^2]$$

E refers to mean




Assume a linear function

$$y(n) = w.x(n)$$

m is replaced by w

Cost function: J

# Cost function

**Observation 1**

1. Cost function is a function of error

**Observation 2**

- Error is [z- f(w,x)]

1. Error is function of expected output (z)
2. Error is function of input (x)
3. Error is function of weights (w)

**Inference**

- Cost function is a function of z, x and w

# Make an adaptive system

- Cost function = function of z,x, &w
- x & z not in our control

Make expected output = predicted output

↓

Minimize error

↓

Minimize cost function

↓

Minimize cost function w.r.t weight

$$J_w(e(n)) = E[(z(n) - f(x(n), \mathbf{w}))^2]$$

$$\frac{\partial J(e(n))}{\partial \mathbf{w}} = 0$$

$$\boxed{\frac{\partial J(e(n))}{\partial \mathbf{w}} = \frac{\partial J(e(n))}{\partial e(n)} \frac{\partial e(n)}{\partial \mathbf{w}} = 0}$$

How error function changes w.r.t to error?

How error changes w.r.t to w?

$$\frac{\partial J(e(n))}{\partial \mathbf{w}} = E\left[\frac{\partial e^2(n)}{\partial e(n)} \frac{\partial e(n)}{\partial \mathbf{w}}\right] = -2E[e(n)\mathbf{x}(n)] = 0$$

# MSE condition – linear system

| x | y (generated) | z (desired or expected) | e = z-y | e.X |
|---|---|---|---|---|
| X1 | Y1 | Z1 | e1 | X1.e1 |
| X2 | Y2 | Z2 | e2 | X2.e2 |
| X3 | Y3 | Z3 | e3 | X3.e3 |
| … | … | … | … | … |
| … | … | … | … | … |
| … | … | … | … | … |
| Xn | Yn | Zn | en | Xn.en |

Expectation of last column should go to zero